

André Carmo

AI Engineer · Automation & Agentic Systems

LLM agents and workflow automation in production · RAG · n8n · Python & TypeScript · Claude and OpenAI APIs · MCP · 30 years in IT

Porto, Portugal · +351 912 405 722 · andre.carmo@gmail.com · linkedin.com/in/stonedge · andrecarmo.pt

Portuguese (native) · English (full professional) · Spanish (full professional) · French (elementary)

PROFILE

AI engineer specialised in automation. Since 2025 I design, build and run LLM agents and workflow automations in production for SMEs and professional practices (healthcare, real estate, legal, wellness): WhatsApp agents that answer, qualify and book 24/7, RAG and legal research assistants over company documents, and integrations between CRMs, calendars, email and messaging through REST APIs and webhooks. Stack: Claude API and Claude Agent SDK, OpenAI API, MCP, tool calling, n8n, Python and TypeScript, Supabase/pgvector, Docker. Behind it, 30 years in IT, including 5 years in software engineering teams at Altran and Capgemini Engineering (frontend development, functional and integration testing) and a decade running infrastructure. That background is what makes the agents production-grade: scenario-based regression tests, guardrails and human-in-the-loop fallbacks on every deployment.

CORE SKILLS

AI & LLM engineering

LLM agents and agentic workflows multi-agent systems tool and function calling
structured outputs prompt engineering
RAG, embeddings and vector databases (pgvector, Supabase) guardrails and human-in-the-loop
LLM evaluation (scenario regression suites, golden cases)
observability (structured logging, token and cost tracking) Claude API Claude Agent SDK
Claude Code MCP (Model Context Protocol) OpenAI API

Automation & integration

n8n (self-hosted, upgrades and custom nodes) workflow automation WhatsApp Business API
REST APIs webhooks OAuth
CRM, ERP, calendar and email integrations (Google Workspace, Microsoft 365) Microsoft 365 Copilot
business process automation for SMEs

Languages & data

Python TypeScript Node.js JavaScript SQL PostgreSQL Supabase MySQL Bash
PowerShell

Platforms & delivery

Docker Linux (VPS) Git, GitHub Azure DevOps JIRA Scrum
AI-assisted development (Claude Code with skills, hooks and subagents)

Quality

functional, integration and regression testing test design pytest
evaluation of non-deterministic systems

Familiar with

LangChain, LangGraph, LlamaIndex, CrewAI (concepts, no production use) Make, Zapier (evaluated)

Background

Angular, React, TypeScript Windows Server, Active Directory, Exchange Hyper-V, VMware VoIP
ERP PHC

PROFESSIONAL EXPERIENCE

AI Automation Engineer (freelance) | Self-employed

Apr 2025 – present

Porto · remote for Portuguese and international clients

- Dental clinic: 24/7 WhatsApp scheduling agent in production with real patients. WhatsApp Business API, n8n orchestration and a Claude API agent with tool calling into Google Calendar and the clinic CRM (REST, webhooks): qualifies the request, checks live availability, books, confirms, sends reminders and manages a waiting list. Designed to cut inbound calls by around 70%.
- Law firms: legal research and drafting assistant in production since November 2025, built from scratch in Python and tailored to each firm. Aggregates several legal codes for research and review, analyses contracts, drafts from structured contract templates that only need the client's data and the current situation, triages email and searches the firm's documents; the lawyer keeps the final word. Cuts a typical eight-hour task to about one hour: 30 minutes assisted by the AI and 30 minutes of lawyer review.
- Real-estate agency: lead qualification agent in production. Scores and qualifies leads from property portals, enriches the CRM automatically and hands qualified leads to the agents with context. Designed to triple qualified leads and save about five hours a day of manual follow-up.
- Aesthetics centre: appointment confirmation, reminder and waiting-list automation in production, available 24/7; designed to reduce no-shows by around 40%.

- RAG assistants for document analysis and technical support in production: embeddings and hybrid retrieval on Supabase/pgvector with metadata filters, answers with sources, escalation to a human with full context when confidence is low.
- Engineering practice: moved conversational logic from visual n8n flows into code with the Claude Agent SDK (Python and TypeScript) after scenario testing exposed unhandled conversation states. Bookings are validated against the real calendar before the user sees them, failures follow defined fallback paths, and a scenario-based regression suite (pytest) runs on every prompt or model change. Structured logging with token and cost tracking per conversation.
- Multi-agent operations system on Claude Code, the Agent SDK and MCP servers (subagents, reusable skills, browser, email and calendar tools) that I run daily; measured token cost per task and redesigned the system to cut it by about two thirds; append-only logs with hash verification and supersede records after a data-loss incident.
- Delivery: discovery with business owners, scoping by impact, first release typically within 48 hours, documentation and handover. Deployments on self-hosted Docker (VPS), managed platforms and client servers, with direct API integration into the clients' CRM and ERP systems.

Functional Tester (mid) and Frontend Developer | Capgemini Engineering

Mar 2021 – Mar 2025

Vila Nova de Gaia

- Functional, integration and regression testing in Scrum teams for Schneider Electric, Keurig and Winbiz: test design, execution and defect management in Azure DevOps and JIRA; SQL; Android and iOS environments; hardware and software flashing and brewer testing (Keurig).
- Post-Quantum Cryptography project (2024–25): research, documentation, code, performance testing and a VPN implementation.
- Frontend development for VW Touareg (2021): Angular, React, TypeScript, Node.js; mobile-oriented MVP in Scrum.

Frontend Developer | Altran

Jan 2020 – Mar 2021

Vila Nova de Gaia

- TUA and VW Hawaii projects: Angular, React, TypeScript, Node.js, HTML5/CSS3; Git, GitHub and GitLab workflows. GLPI project: MariaDB.

IT Consultant | Altran / IEF Academy

Jun 2019 – Dec 2019

Vila Nova de Gaia

- Windows Server 2003–2012, Active Directory, Azure AD, Exchange, DNS, SharePoint, PowerShell, Hyper-V and VMware; Scrum.

Technical Director and IT Engineer | Diodo

Mar 2006 – Jan 2012

Gondomar

- Infrastructure, Active Directory and Exchange, virtualisation (Hyper-V, VMware), storage and backups, security policies, ERP PHC, SQL and MySQL, Unix-to-SQL migrations, PBX and GSM integration into VoIP; trained and led the technician team.
- As technical director (2011–12): import and export sales, public-tender platforms and documentation, technical quality reports with LABELEC engineers.

IT Project Manager, IT Consultant and Computer Technician | Diodo · Stonedge World · sole trader

Oct 1996 – Mar 2006

Gondomar

- Web store integrated with ERP PHC (project 40% funded by IAPMEI, QREN 2006); enterprise Wi-Fi, network and telecom projects; hardware and software maintenance; client management and team training.

EDUCATION

Engineer's Degree (5-year, pre-Bologna), Electronics and Telecommunications · Universidade de Aveiro, 2004. Final-year project at Eindhoven University of Technology (TU/e), Netherlands, 2002, the basis of the publication below.

Mechatronics, Robotics, Control and Automation Engineering · CINEL, 2016

CERTIFICATIONS AND PUBLICATION

TypeScript Essential Training · Learning TypeScript · JavaScript: Async · SQL Language · Learning AngularJS 2

Publication: A Study on the Transformation of Sequential Streaming Algorithms into Parallel YAPI Programs